Adaptive Advance-Time Control for nFAPI Split under Unpredictable Latency

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Abstract—

Index Terms—nFAPI, O-RAN, Functional Split, Adaptive Control.

I. Introduction

A. Introduction

In recent years, the Open Radio Access Network (O-RAN) architecture has become popular. O-RAN allows for more flexible network deployments. The Network Functional Application Platform Interface (nFAPI) is an important part of this architecture. nFAPI helps split the O-RAN Distributed Unit (O-DU) into O-DU High and O-DU Low parts. According to the nFAPI specification [1], the Virtual Network Function (VNF) includes Layer 2 and Layer 3 operations. The Physical Network Function (PNF) handles the Layer 1 Physical layer processing. This functional split physically separates the Media Access Control (MAC) layer and the Physical (PHY) layer. Crucially, the MAC scheduler responsible for allocating radio resources resides within the VNF. Figure 1 shows this O-RAN nFAPI functional split architecture and the separation of these components.

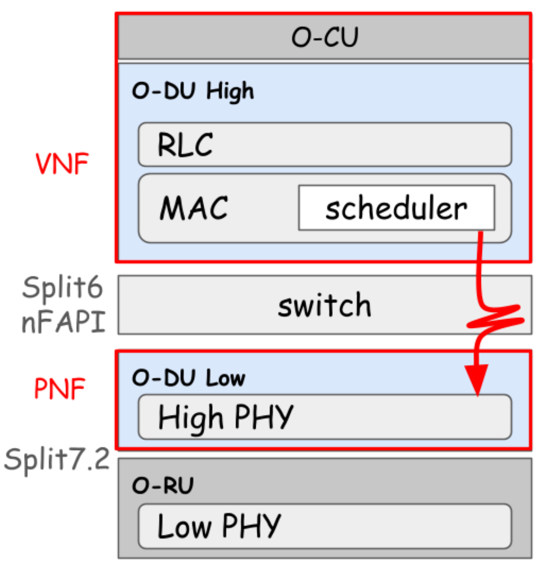

Fig. 1. O-RAN nFAPI functional split architecture.

Deploying the VNF and PNF on different machines introduces significant challenges. Because the MAC scheduler is separated from the PHY layer, strict slot synchronization is required between these distributed components. However, even when deployed in the exact same network environment, different traffic loads directly impact the latency. Figure 2 illustrates the latency between the VNF scheduler and the PNF under varying traffic loads. As traffic increases, three factors heighten the overall delay: the scheduler requires more processing time, the data packing time lengthens, and the VNF-to-PNF One-Way Delay (OWD) is affected. Consequently, the VNF cannot reliably transmit data exactly at the slot boundary. Instead, it must dynamically advance its scheduling point to compensate for these varying delays and ensure packets arrive at the PNF on time.

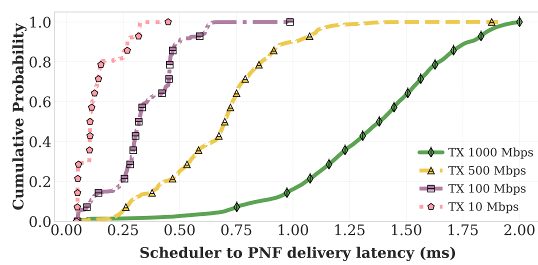

Fig. 2. Observation of latency between VNF scheduler and PNF under different traffic loads.

To lower deployment costs, operators often want to deploy Split-6 nFAPI gNBs over non-ideal networks. However, beyond traffic load variations, these non-ideal networks introduce additional, unpredictable latency. Figure 3 shows the latency jitter caused by intermediate routers and switches in a non-ideal deployment. Existing open-source solutions, such as OpenAirInterface (OAI), provide basic nFAPI support but were not designed to fully comply with the nFAPI specification's delay management procedures [1]. Therefore, they suffer from synchronization failures when encountering unpredictable latency from routers and switches. To address these challenges, this paper proposes a dynamic trigger scheduler adjustment algorithm that fully complies with the specification. Through systematic experiments, we verify the effectiveness of this dynamic adjustment and its benefits to overall network reliability.

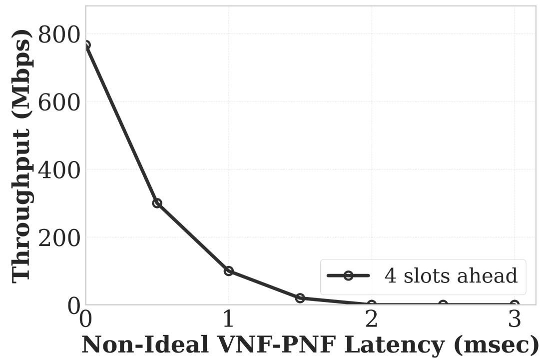

Fig. 3. Impact of unpredictable network latency in a non-ideal environment.

B. Related Works

C. Contributions

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

II. System Model/Architecture

TABLE I
nFAPI Delay Management Configuration Parameters

Parameter	Type	Description
timingWindow	uint16_t	Timing window for delay management. See section 3.2.9 of [10]
timingMode	uint8_t	Timing mode for Timing Info. See section 3.2.9 of [10]
timingPeriod	uint8_t	Timing period for Timing Info. See section 3.2.9 of [10]

The following assumptions were made in this paper.

- Assumption 1.
- Assumption 2.
- Assumption 3.

III. Proposed Method

IV. Numerical/Simulation/Experimental Results

Computer simulations/Experiments were conducted to verify xxx (the effectiveness of the proposed analytical model). In the following figures, each point represented the average value of 10^5 samples. In each sample, we collect the statistics of xxx. Symbols and lines were used to show the simulation and analytical results, respectively.

XXX scenarios were investigated. Scenario I was designed to demonstrate/verify the accuracy/effectiveness of Contribution 1.

A. Contribution 1

Fig. 1.1: Purpose X axis: Y axis:

B. Contribution 2

Fig. 2.1: X axis: Y axis:

C. Contribution 3

Fig. 3.1: X axis: Y axis:

References

[1] Small Cell Forum, “5g network functional api (nfapi) specification,” Small Cell Forum, Tech. Rep. SCF225, 2021. [Online]. Available: https://www.smallcellforum.org/